Companion XLVII: The Bispectrum in the Relaxation-Driven Cyclic Cosmology

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Abstract

The bispectrum provides a powerful probe of non-Gaussianity and mode coupling in the large-scale structure of the Universe. In the standard cosmological model, non-Gaussianity arises from non-linear gravitational evolution and baryonic processes, while primordial non-Gaussianity is typically assumed to be small. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential introduces a new source of non-Gaussianity, as the relaxation mechanism modifies the temporal evolution of density perturbations in a scale-dependent manner. This Companion develops a continuous description of the bispectrum in RDCC, deriving how the relaxation scale influences the shape, amplitude, and configuration dependence of the bispectrum. The resulting framework provides a distinctive observational signature of the RDCC gravitational field.

1 Introduction

The bispectrum is the Fourier transform of the three-point correlation function and provides a direct measure of non-Gaussianity in the matter distribution. While the power spectrum captures the variance of density fluctuations, the bispectrum captures the coupling between modes of different scales. In the standard cosmological model, this coupling arises from non-linear gravitational evolution and baryonic processes.

In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the coupling between modes in a scale-dependent manner, producing a distinctive bispectrum signature that reflects the relaxation scale $k_{\rm rel}$.

2 The RDCC Density Field and Mode Coupling

The density contrast evolves according to

$$\delta(\mathbf{k},a) = D(a)\,\delta(\mathbf{k},a_i) + \int d^3q\, F_2(\mathbf{q},\mathbf{k}-\mathbf{q},a)\,\delta(\mathbf{q})\,\delta(\mathbf{k}-\mathbf{q}) + \cdots\,, \tag{1}$$

where F_2 is the second-order kernel.

In RDCC, the kernel becomes scale-dependent:

$$F_{2,\rm RDCC} = F_2\left[1-\chi_F\left(\frac{k}{k_{\rm rel}}\right)^{\alpha}\right]. \tag{2}$$

This modification alters the coupling between modes and introduces a new source of non-Gaussianity.

3 The RDCC Bispectrum

The bispectrum is defined as

$$\langle \delta(\mathbf{k}_1)\delta(\mathbf{k}_2)\delta(\mathbf{k}_3)\rangle = (2\pi)^3 \delta_D(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) B(k_1, k_2, k_3). \tag{3}$$

In RDCC, the bispectrum becomes

$$B_{\rm RDCC} = 2F_{2,\rm RDCC}(k_1, k_2) P_{\rm RDCC}(k_1) P_{\rm RDCC}(k_2) + \text{cyc.}, \tag{4}$$

where the RDCC power spectrum is

$$P_{\rm RDCC}(k) = P(k) \left[1 - \chi_P \left(\frac{k}{k_{\rm rel}}\right)^{\alpha}\right]. \tag{5}$$

The bispectrum therefore acquires a scale-dependent modification that reflects the relaxation scale.

4 Shape Dependence and RDCC Signatures

The bispectrum depends on the shape of the triangle formed by the three wave vectors. In the standard cosmological model, the bispectrum is largest for squeezed, equilateral, and folded configurations.

In RDCC, the suppression of small-scale modes reduces the amplitude of the bispectrum in equilateral configurations, while the enhanced coherence of large-scale modes increases the amplitude in squeezed configurations.

The RDCC bispectrum therefore exhibits a characteristic enhancement in squeezed triangles, providing a direct observational probe of the relaxation scale.

5 The Reduced Bispectrum and Observational Probes

The reduced bispectrum is defined as

$$Q(k_1, k_2, k_3) = \frac{B(k_1, k_2, k_3)}{P(k_1)P(k_2) + P(k_2)P(k_3) + P(k_3)P(k_1)}. \tag{6}$$

In RDCC, this becomes

$$Q_{\rm RDCC} = Q \left[1 - \chi_Q \left(\frac{k}{k_{\rm rel}}\right)^{\alpha}\right]. \tag{7}$$

This modification produces a distinctive scale dependence that can be measured in galaxy surveys, weak lensing surveys, and CMB lensing reconstructions.

6 Conclusion

The bispectrum in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the mode-coupling kernel, the power spectrum, and the shape dependence of the bispectrum, producing distinctive signatures in squeezed, equilateral, and folded configurations. These effects provide a direct observational probe of the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of non-Gaussianity, mode coupling, and the higher-order statistics of large-scale structure in RDCC.

References

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[4] Companion XXXI: RDCC Halo Model and Non-Linear Structure.

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